

LLMs Note-Taking Framework

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1 Summary

This week we looked into Large Language Models (LLMs). These are chat bots such as chatGPT and others. We looked at how they are trained and the behind the scenes of how they determine/predict the following words in sentences and prompts. Large Language Models are comprised of transformers. Transformers are extremely important to the training of the models for a couple of reasons, speedup due to parallelization, and contextual understanding via attention. Transformers allow parallelization because of how they are structured and they can process the vectors for each word/token in a sentence at the same time (Transformers, the tech behind LLMs — Deep Learning Chapter 5). Perhaps the most impactful part of transformers is attention. Attention allows the model to add context to a token's vector based off the vector of the words around it in the data (Attention in transformers, step-by-step — Deep Learning Chapter 6). The emergence of transformers made LLMs much more effective and allowed for faster training.

We also looked at the training process for LLMs. The process is broken down into 3 stages: Generative Pre-Training, Supervised Fine Tuning, and Reinforcement Learning from human feedback. In the Generative Pre-training stage the model is given a massive corpus of text and tasked with trying to predict the next token in a sequence. Then the model is further trained in the Supervised Fine Tuning stage. In this stage humans provide example of what the model should be outputting. Then the final stage is Reinforcement Learning from human feedback in which the model creates output that is graded by humans. These grades/ranks are used to create a reward model, the model tries to achieve the highest reward possible. Each of these stages builds off the previous one, initializing the weights with the weights from the final state of the previous stage (How ChatGPT is Trained).

2 Attention Grabbers

1. The first video talked about the second phase of training, reinforcement learning with human feedback. This caught my eye because I actually did this a couple years ago as a side gig.

2. In the second video, it shows a visualization of embeddings in a 3D space and it showed an example of the $\text{Queen} = \text{King} - \text{man} + \text{woman}$. I understood the concept beforehand, but this visualization made it make even more sense.
3. I think the concept of temperature within the Softmax function is really interesting. Getting the model to choose a less likely word to make the sentence/paragraph flow more naturally is not something that I would have thought of.
4. I thought it was interesting how when training an LLM, humans act as both the user and the chatbot in the supervised learning stage. It's like giving a kid an example of how to do something.
5. 2 Trillion!! That's how many parameters the Llama 4 Behemoth uses. That's just insane, I never would've thought there were that many parameters behind the scenes, a few million maybe, but trillions is unfathomable.

3 Questions

1. The videos mention adding one word at a time to the model input to have the model predict the next word, then the next word, until its done. Is this how modern models or does it do multiple words or sentences at a time?
2. The videos say for a person to read all the data that LLMs are trained on would take a couple thousand years, how do developers find and filter out bad data like the counting subreddit?
3. What's the biggest difference and benefits of doing both supervised fine tuning and reinforcement learning from human feedback?
4. Do you think we could ever train a model so well that it could tell us how/why it updated/manipulated it's weights? Describe its own learning process in a way?
5. Is there a limit to the number of dimensions you could add to embeddings? Is it strictly computational or are there other limiting factors?